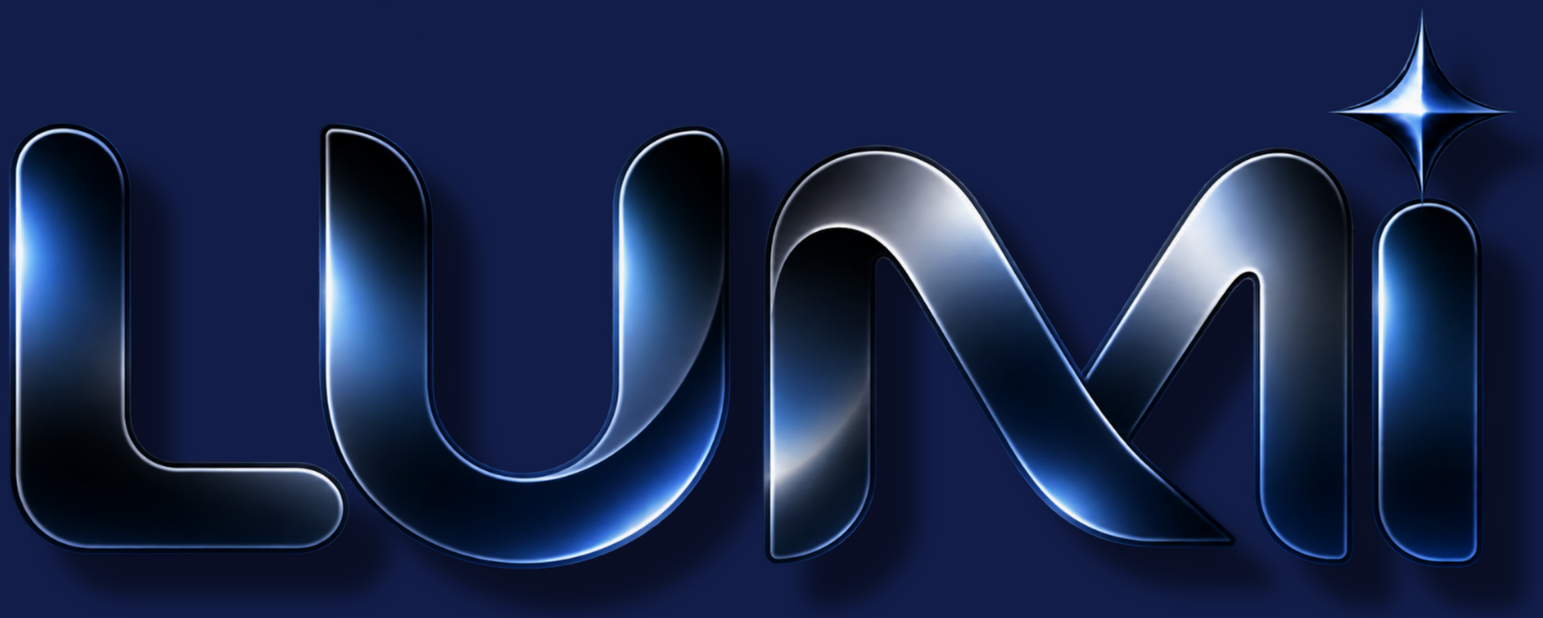

Long&short time User Memory with Image selection

26-2 Human-Compuer Interaction
Team4 잘생긴팀

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OUR PLAN

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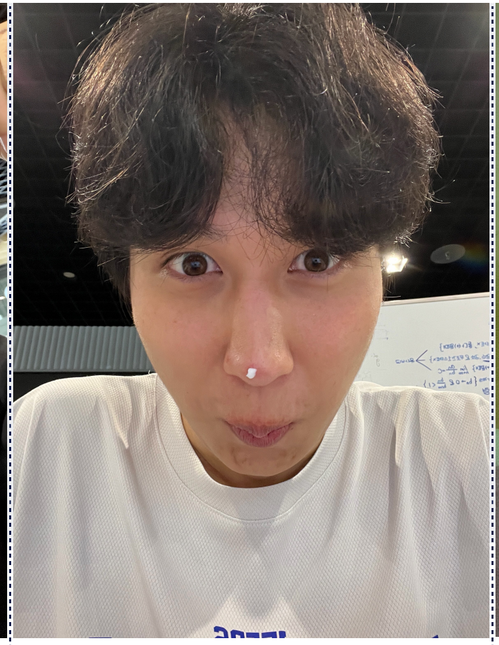

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PROBLEM

LLM

ONLINE STORE

MUSINSA

29CM

ABLY

We experienced few problems using LLM & Online Store

1. Difficulty in translating
   User Taste -> Fashion Words

2. Systems can't distinguish between
   Users' original preference vs Situational needs

   ex) Searching for formal style according to interview next week, **system suggests formal style items after this session.**

PROJECT INTRODUCTION

OVERVIEW

- LLM Agent **Interpret User's Taste** with imgae selection and text question
- Seperate Long and Short-time memory to **divide original preference and situational needs**

OBJECTIVE

- Support Taste Exploration
- Long & Short time memory
- Dividing Confirmed preference and Hypothesis
- User Control

RESEARCH QUESTION

- Does **combining image and text selection** improves final recommendation and **users' understanding** of there preference?
- Does **separating long and short-time memory** with user feedbacks improves shopping experience?

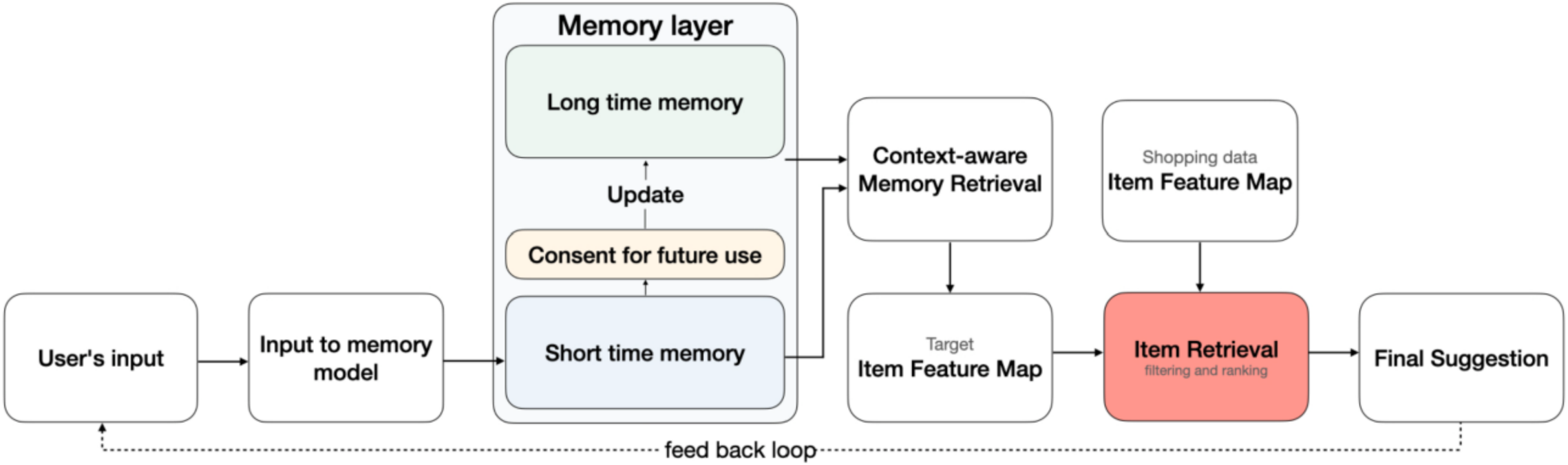
OUR SUGGESTION IS...

LUMI

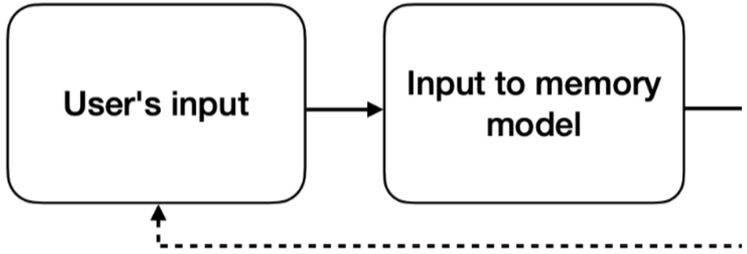
Long&short time User Memory with Image selection

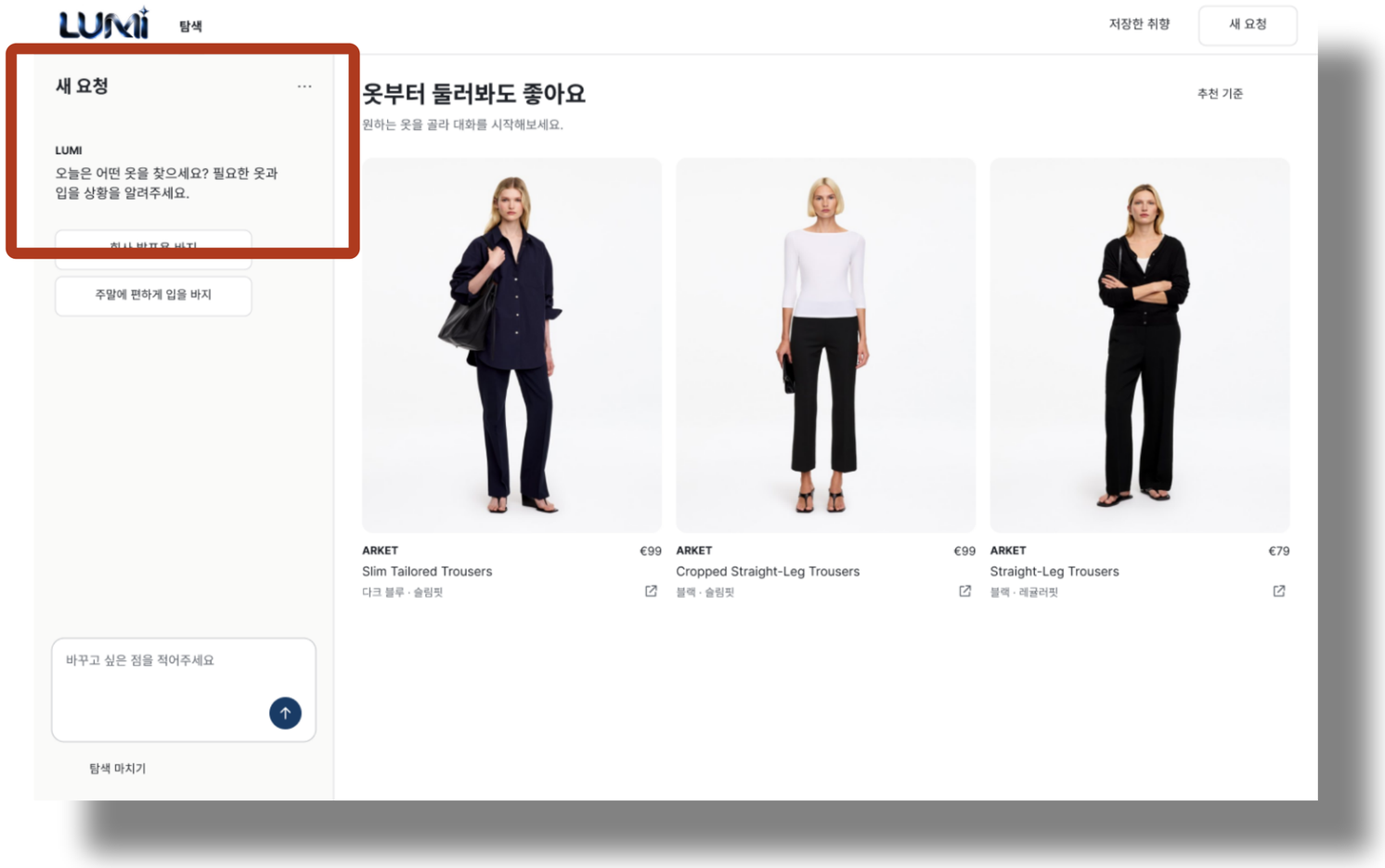
OUR SYSTEM 'LUMI'

SYSTEM PIPELINE



OUR SYSTEM 'LUMI'

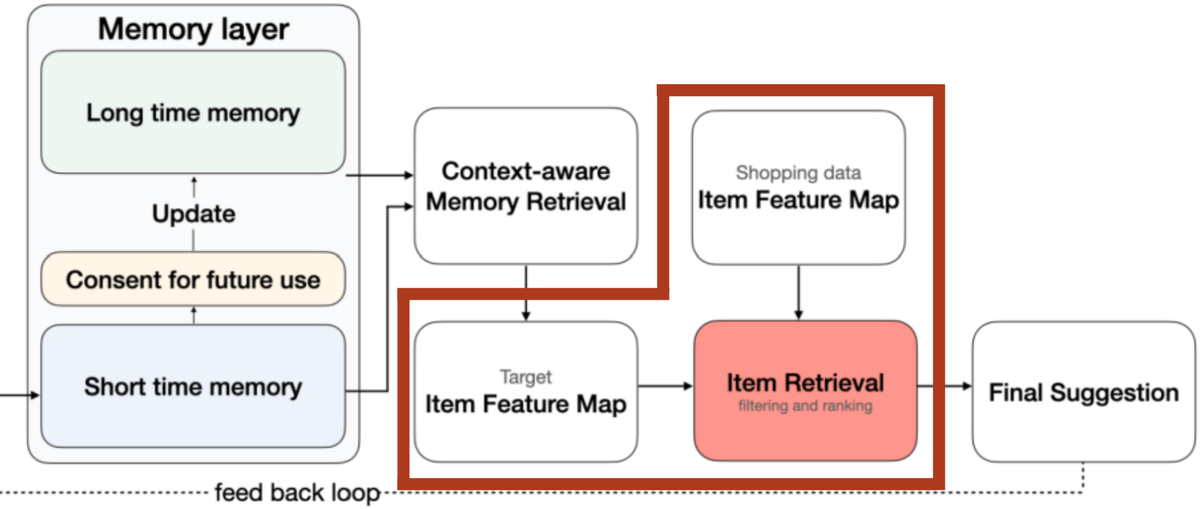


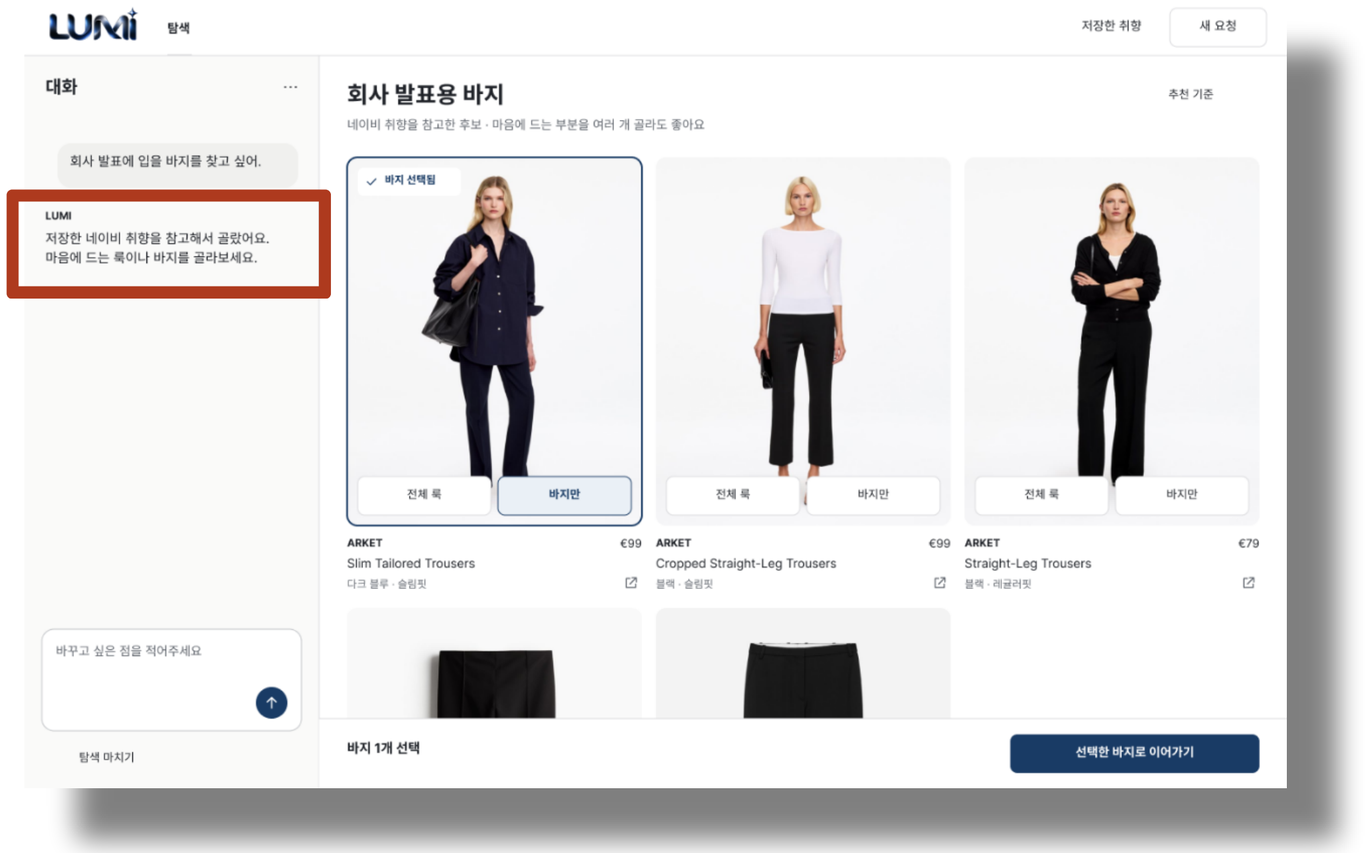


USER INPUT

- User types needs to system
- This input is given to memory model

OUR SYSTEM 'LUMI'





VLM

- Qwen Family
- CLIP & FashionCLIP
- SIGLIP2

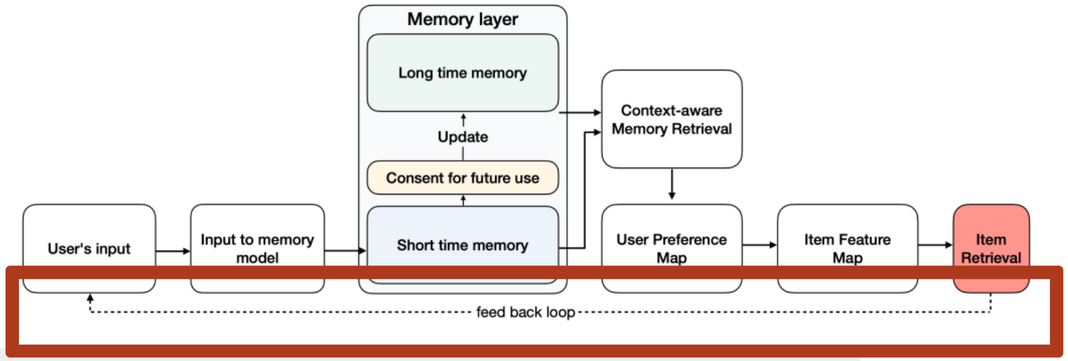
FEATURE MAP

- Made by our team
- Detailed attributes in fashion domain

INITIAL RECOMMENDATION

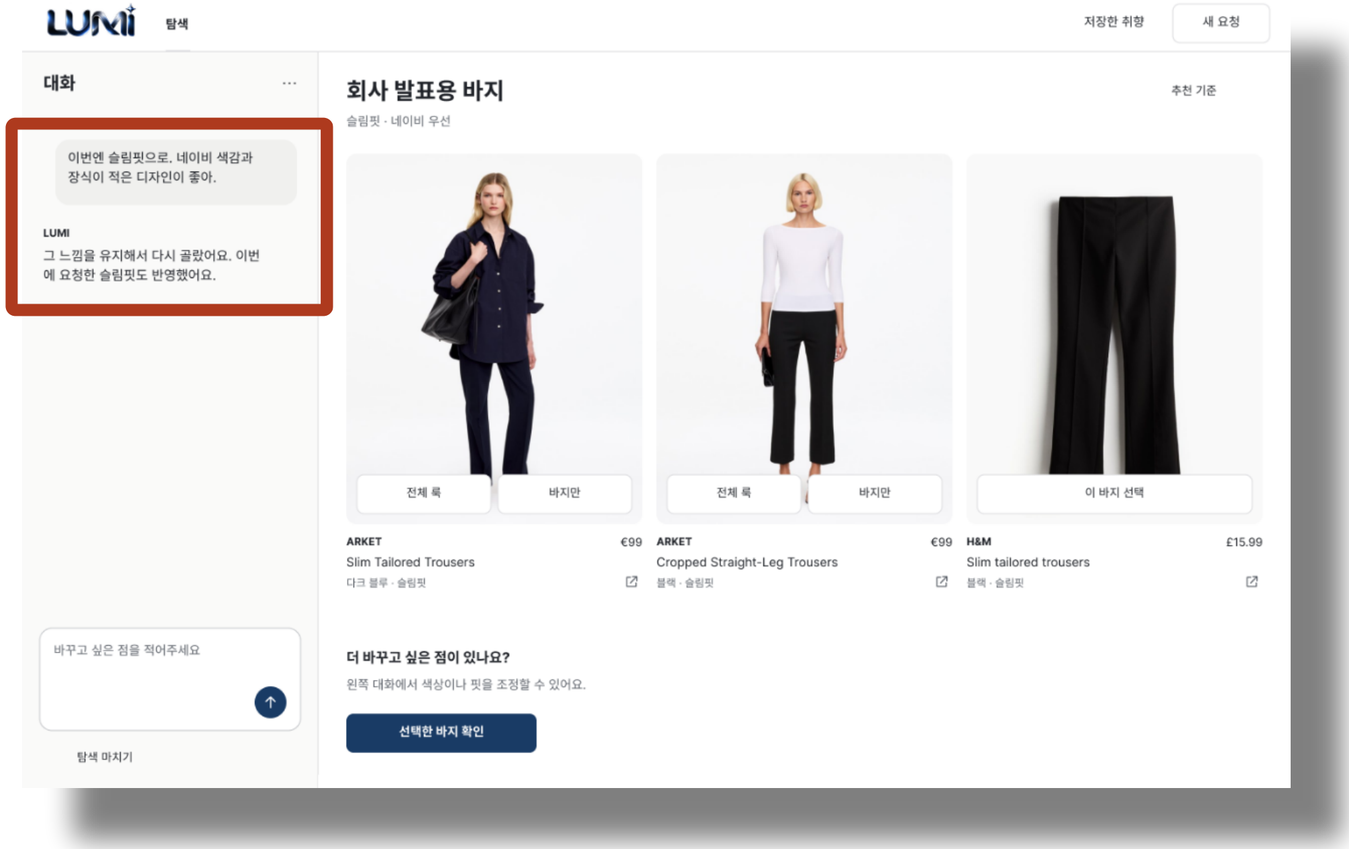
- User Input is treated as short-time memory
- Original preference is saved in long-time memory
- System gives initial recommendation considering both

OUR SYSTEM 'LUMI'



STOP CONDITION

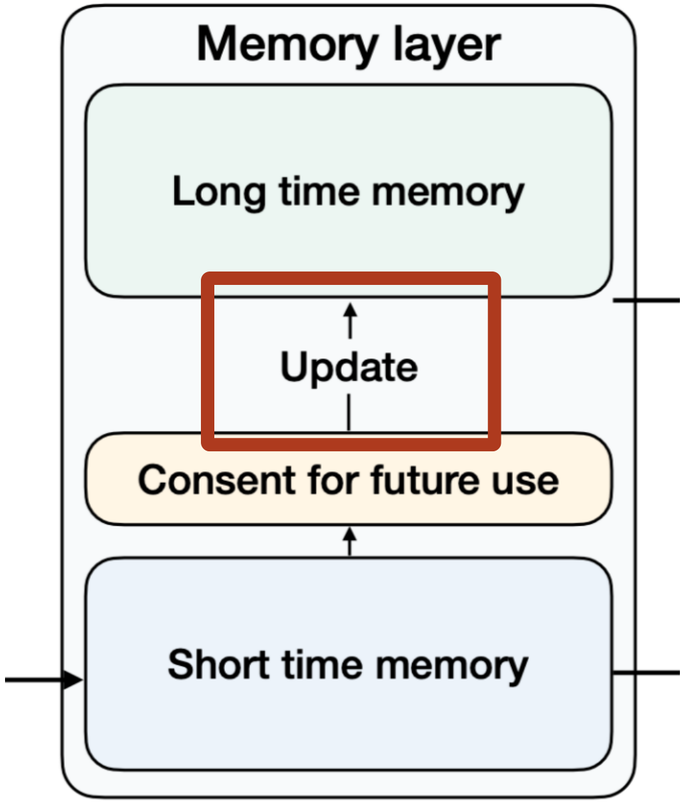
- Until User makes final selection
- If LLM depends sufficient preferences are found, system notice it to user

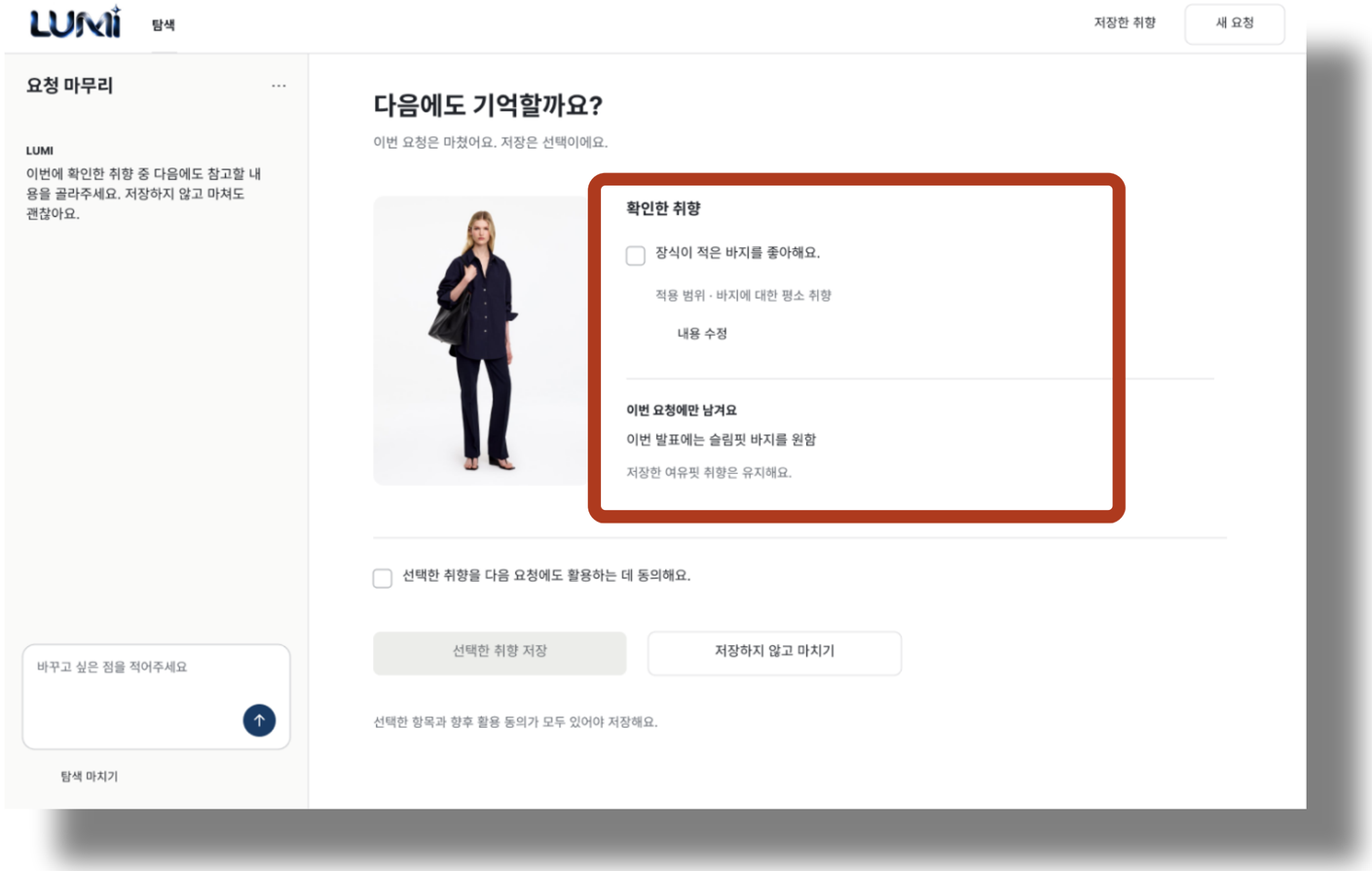


FEEDBACK LOOP

- LLM interpret the purpose of image selection **(Hypothesis)**
- LLM **asks question** about user purpose based on hypothesis
- User can give **feedback** and System apply it as **Confirmed Preference**

OUR SYSTEM 'LUMI'

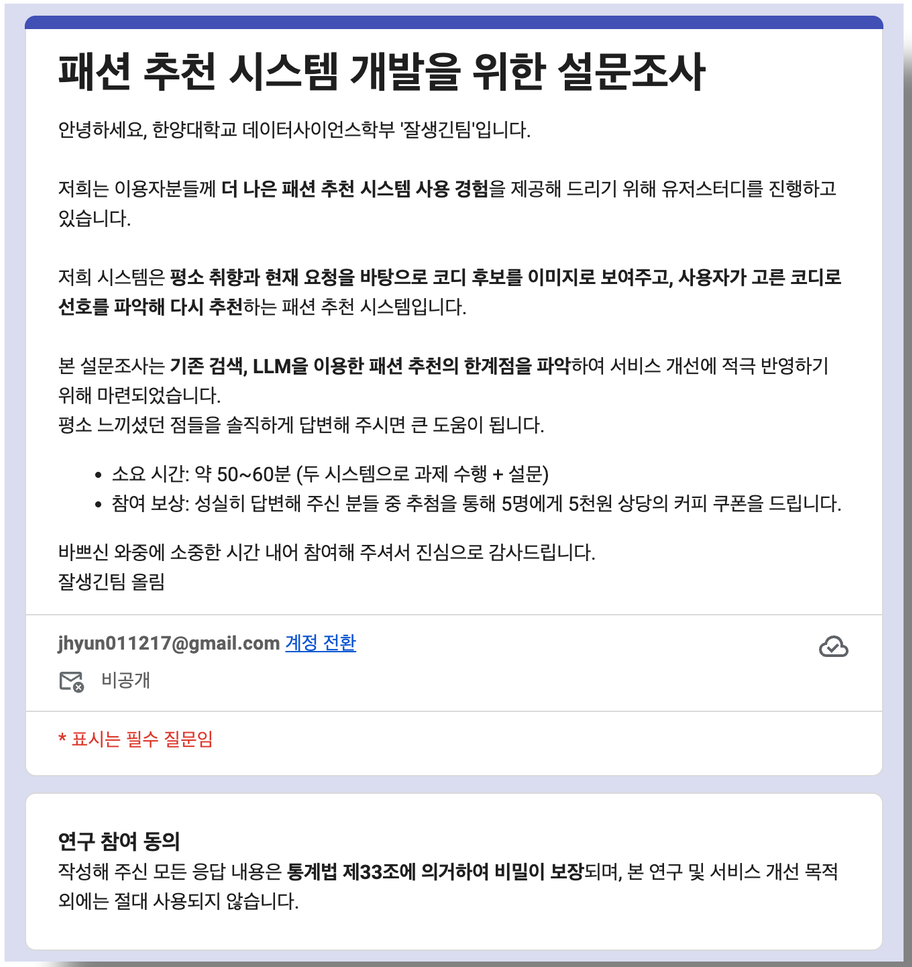




MEMORY UPDATE

- After user's final selection, system summarize newly found user's preference in this session
- User can choose whether it becomes to original preference (Short time memory → Long time memory)

USER STUDY



패션 추천 시스템 개발을 위한 설문조사

안녕하세요, 한양대학교 데이터사이언스학부 '잘생긴팀'입니다.

저희는 이용자분들께 **더 나은 패션 추천 시스템 사용 경험**을 제공해 드리기 위해 유저스터디를 진행하고 있습니다.

저희 시스템은 **평소 취향과 현재 요청을 바탕으로 코디 후보를 이미지로 보여주고, 사용자가 고른 코디로 선호를 파악해 다시 추천**하는 패션 추천 시스템입니다.

본 설문조사는 **기존 검색, LLM을 이용한 패션 추천의 한계점을 파악**하여 서비스 개선에 적극 반영하기 위해 마련되었습니다.
평소 느끼셨던 점들을 솔직하게 답변해 주시면 큰 도움이 됩니다.

- 소요 시간: 약 50~60분 (두 시스템으로 과제 수행 + 설문)
- 참여 보상: 성실히 답변해 주신 분들 중 추첨을 통해 5명에게 5천원 상당의 커피 쿠폰을 드립니다.

바쁘신 와중에 소중한 시간 내어 참여해 주셔서 진심으로 감사드립니다.
잘생긴팀 올림

jhyun011217@gmail.com 계정 전환

비공개

* 표시는 필수 질문임

연구 참여 동의
작성해 주신 모든 응답 내용은 **통계법 제33조에 의거하여 비밀이 보장**되며, 본 연구 및 서비스 개선 목적 외에는 절대 사용되지 않습니다.

USER STUDY DESIGN

- 20~24 Participants (Various Age, Gender)
- Baseline(GPT+Musinsa) VS LUMI prototype
- About 40~50 minutes
  Pre-Survey → Baseline, LUMI use → Post-Survey

SURVEY STRUCTURE

- Pre-survey: User information
  (Including fashion knowledge & shopping experience)
- Post-survey: Comparing baseline and LUMI
- SUS & NASA-TLX (On-Paper) : Usability and Workload

2 REQUEST FOR EACH BASLINE & LUMI

1. To check for Long-time memory
2. Give situation which is different with usual context

USER STUDY - STATISTICAL ANALYSIS

WHAT WE MEASURE

- RQ1: Control & Appropriateness
- RQ2: Expression & Understanding
- SUS: Usability
- NASA-TLX: Workload

HOW WE ANALYZE

- RQ1: Wilcoxon Signed-Rank Test
- RQ2: One-Sample Wilcoxon Test
- SUS: Holm Correction (α = .05)
- NASA-TLX: Order Effect & Correlation

OUR GOAL IS

To evaluate whether **LUMI improves users' control, preference expression, and outfit selection experience** compared to the baseline with **agreeable workload**

FUTURE WORK & NEXT STEPS

IMPLEMENT

- Feature Extraction Model (Data Crawling, Annotation)
- Memory Layer
- LLM, Agent tuning

USER STUDY

- With Prototype
- Statistical Analysis

PROTOTYPE

- Combine back-end & front-end
- All pipeline into one system

ENHANCMENT

- Based on User Study
- Higher Accuracy of LLM's hypothesis generation

LUMI

A personalized Recommendation Agent that **Distinguishes and Reflects Long-term tastes and Short-term preference dynamics**

Thank You

JAEHYUN KIM, KYEONHO KIM, JONGHYUN NOH, SEYONG PARK, JUNSANG LEE

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